

where $\mathbf{F}_n = \mathbf{F}(\mathbf{q}_n) = -\nabla U(\mathbf{q}_n)$. The force computed at the end of a given step may be reused at the start of the following step, thus effectively a single force evaluation is needed at each timestep (the method has a similar “cost” to the Euler method, if we measure cost in terms of force evaluations). The derivation of the Störmer form from the velocity Verlet form is straightforward: write down two consecutive steps of (2.5)–(2.7) then eliminate velocities.

The method is widely used but is often stated in other forms, so let us consider these here.

The most straightforward rewriting of the Verlet method is to put the equations and the discretization in Hamiltonian form, i.e. introducing momenta $\mathbf{p} = \mathbf{M}\mathbf{v}$, and thus $\mathbf{p}_n = \mathbf{M}\mathbf{v}_n$, which results in the flow map approximation (taking us from any point in phase space $(\mathbf{q}, \mathbf{p})$ to a new point $(\mathbf{Q}, \mathbf{P})$):

$$\mathbf{Q} = \mathbf{q} + h\mathbf{M}^{-1}\mathbf{p} + \frac{h^2}{2}\mathbf{M}^{-1}\mathbf{F}(\mathbf{q}), \quad \mathbf{P} = \mathbf{p} + \frac{h}{2}[\mathbf{F}(\mathbf{q}) + \mathbf{F}(\mathbf{Q})]. \quad (2.8)$$

Alternatively,

$$\mathbf{p}_{n+1/2} = \mathbf{p}_n + (h/2)\mathbf{F}_n, \quad \mathbf{q}_{n+1} = \mathbf{q}_n + h\mathbf{M}^{-1}\mathbf{p}_{n+1/2}, \quad \mathbf{p}_{n+1} = \mathbf{p}_{n+1/2} + (h/2)\mathbf{F}_{n+1}.$$

Returning to (2.5)–(2.7), write out the formulas for two successive steps $((\mathbf{q}_{n-1}, \mathbf{v}_{n-1}) \mapsto (\mathbf{q}_n, \mathbf{v}_n))$ and $(\mathbf{q}_n, \mathbf{v}_n) \mapsto (\mathbf{q}_{n+1}, \mathbf{v}_{n+1}))$ and note that, from

$$\mathbf{v}_n = \mathbf{v}_{n-1/2} + (h/2)\mathbf{M}^{-1}\mathbf{F}_n,$$

one has

$$\begin{aligned} \mathbf{v}_{n+1/2} &= \mathbf{v}_{n-1/2} + h\mathbf{M}^{-1}\mathbf{F}_n, \\ \mathbf{q}_{n+1} &= \mathbf{q}_n + h\mathbf{v}_{n+1/2}. \end{aligned}$$

How do we use this if, as is typical, initial conditions of the form $\mathbf{q}(0) = \mathbf{q}_0, \mathbf{v}(0) = \mathbf{v}_0$ are specified? It is necessary to define an initialization procedure for $\mathbf{v}_{-1/2}$:

$$\mathbf{v}_{-1/2} = \mathbf{v}_0 - (h/2)\mathbf{M}^{-1}\mathbf{F}(\mathbf{q}_0).$$

And it is also necessary to use, at any subsequent step,

$$\mathbf{v}_n = \mathbf{v}_{n-1/2} + (h/2)\mathbf{M}^{-1}\mathbf{F}_n,$$

if $\mathbf{q}_n, \mathbf{v}_n$ are both needed, e.g. for the evaluation of the energy or other velocity-dependent observable.

The difference between formulations is typically subtle, and largely influence the details of computer software design. For many applications the difference can